Polymarket-Conditioned Equity Returns: A Polarity-Aware Empirical Study

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Abstract

We test whether equities linked to Polymarket questions exhibit positive average net return after a probability-based eligibility signal and before the associated event deadline. The object of interest is a conditional expectation, not a causal information-diffusion mechanism. The raw archive contains 1,293 question-asset candidate rows from August 2024 through June 2026. Exact duplicate collapse and a targeted semantic audit produce a primary universe of 1,182 candidates with explicit contract polarity. A fixed-notional H1 evaluation yields 887 tradable observations with mean net return +1.208% and median +0.188%. The candidate mean remains positive when resampling economic events, but confidence intervals cross zero under symbol and calendar dependence, and no interval excludes zero once each trade is compared against a matched, equal-notional SPY or sector-ETF position over the identical dates. Equal-weight economic events have mean +0.631%, and removing the largest one percent of events reduces this estimate to +0.029%. Geopolitical observations carry the result; earnings observations are approximately centered at zero. We then evaluate a secondary Cross-Entropy Method (CEM) portfolio implementation with four treatments: friction-aware fitness, label-complete walk-forward fitting, bounded half-Kelly sizing, and event-priority allocation. Across ten seeds and four CEM budgets, median excess return over the single January–June 2026 test period is positive on both benchmarks for every budget; however, in seed-paired comparisons the four treatments do not outperform the plain CEM baseline. The evidence supports Polymarket probabilities as continuous, timestamped crowd-belief variables for candidate identification; it does not establish market-relative excess return, identify why the conditional return exists, or support a production-level profitability claim.

CCS Concepts

• **Applied computing** → **Economics**; • **Computing methodologies** → *Model development and analysis.*

Keywords

prediction markets, event-driven equities, conditional expectation, semantic polarity, walk-forward backtesting

1 Introduction

Prediction markets provide continuously revised prices for discrete future events. Unlike a conventional news archive, a market price supplies a time-indexed state variable: the current crowd-implied probability of a contract resolving YES. This creates a practical link between event identification and equity timing when a contract can be mapped to an economically exposed asset. The link is not automatic. Contract wording, outcome direction, event deadlines, duplicated questions, and non-equity resolution rules must be interpreted before the probability can be used as a bullish or bearish signal.

This paper studies the following empirical question:

> Within a Polymarket-identified and semantically cleaned opportunity set, is the average net equity return positive between probability-based signal eligibility and the final eligible pre-event close?

The associated estimand is deliberately narrower than abnormal return, alpha, or slow information diffusion. It asks whether the observed conditional mean is positive in the opportunity set generated by the system. A positive estimate can arise from several mechanisms, including risk compensation, common-news exposure, event selection, or delayed incorporation of information. Mechanism identification is outside the present test.

The paper makes five empirical contributions. First, it defines contract polarity explicitly, so a favorable NO outcome is not mechanically interpreted as a bearish probability signal. Second, it applies deterministic duplicate collapse and a targeted semantic audit before estimating H1. Third, it reports the expectation at candidate, symbol-day, and economic-event levels with cluster and block resampling, and separately against matched equal-notional benchmark trades. Fourth, it states the eligibility-rule provenance exactly: the entry thresholds are outputs of the portfolio optimizer's walk-forward schedule, so the full-sample H1 estimate is descriptive rather than an independent confirmatory test. Fifth, it demonstrates one portfolio implementation and evaluates optimizer variability across ten random seeds and four CEM budgets, including seed-paired treatment-versus-baseline comparisons. The H1 estimate and the CEM portfolio are kept separate: H1 describes the opportunity set; CEM tests whether a constrained implementation can use it.

2 Related Work

Prediction-market prices are commonly interpreted as aggregators of dispersed beliefs [1, 4]. Their empirical forecasting value is documented in both public and corporate settings [5, 7]. However, market prices need not be calibrated probabilities in every contract: liquidity, participant composition, resolution wording, and correlated information sources affect interpretation.

The efficient-market benchmark predicts rapid incorporation of public information into asset prices [2], while costly information acquisition and trading frictions permit incomplete adjustment [3]. Delayed equity responses are documented in settings such as post-earnings-announcement drift [8], and slow-moving capital supplies one possible frictional channel [6]. These literatures motivate the empirical design, but the present data do not distinguish delayed diffusion from alternative explanations. We therefore treat Polymarket as a continuous belief measurement system rather than assume a particular causal mechanism.

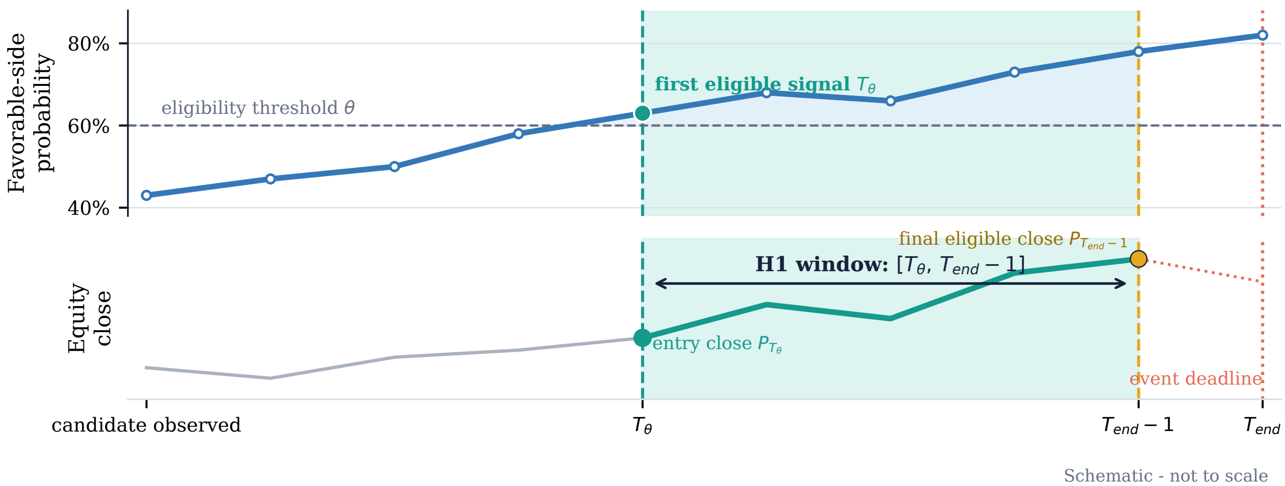

Figure 1: The H1 estimand and its chronology. T_θ is the first point-in-time eligibility decision. H1 measures the mapped equity from that entry close through the last available close before the known event deadline; it does not impose an intervening take-profit or stop-loss. The paths are schematic and not data.

3 Data and Candidate Construction

3.1 Observation unit and chronology

Each source row links one binary Polymarket contract to one U.S.-listed equity or exchange-traded fund. The stored data include the contract question, market and event identifiers, probability path, mapped symbol, scheduled endpoint, daily equity closes, and features available when the candidate is evaluated. The archive spans August 2024 through June 2026. Daily decisions are used because a complete historical hourly equity panel is not available for every symbol; the live implementation can update hourly when current data exist.

Let $p_{i,t}$ denote the probability assigned to the economically favorable side of candidate i. Screening time $T_{\theta,i}$ is the first time this probability crosses a fixed 0.55 screening threshold; it is set at candidate construction and is not optimized. Entry then requires acceptance by the eligibility policy of Section 4.2: the favorable probability must reach a strong threshold immediately, or hold above a floor threshold for a required number of consecutive observations, with caps on the prior probability surge and on the pre-entry equity run-up. The entry close is the first stored equity close on or after acceptance. The endpoint $T_{e,i}$ is the scheduled contract deadline available in the data. H1 exits at the final stored equity close strictly before that deadline. The holding interval is therefore $[T_{\theta,i}, T_{e,i} - 1]$ rather than an early take-profit or stop-loss path, as summarized in Figure 1.

3.2 Polarity

Raw Polymarket data record the probability of YES. That is insufficient for an equity mapping because YES is not universally favorable to the mapped asset. We assign

$$s_i \in \{-1, 0, +1\}, \qquad p^+_{i,t} = \begin{cases} p_{i,t}, & s_i = +1, \\ 1 - p_{i,t}, & s_i = -1, \end{cases}$$

where $s_i = +1$ denotes YES-bullish, $s_i = -1$ denotes NO-bullish, and $s_i = 0$ denotes no stable long-equity interpretation. Deterministic rules cover simple contract classes such as single-company regulatory approval. More complex questions use the audited semantic label. Polarity is an ontology decision: it states which outcome is favorable. It is not itself a learned return forecast.

The cleaned primary universe contains 1,071 YES-bullish candidates, 99 NO-bullish candidates, and 12 candidates without a usable side. Among valid H1 observations, 844 are YES-bullish and 43 are NO-bullish. This imbalance is important when interpreting the polarity ablation in Section 5.

3.3 Duplicate collapse and semantic audit

Exact source duplicates are removed globally before any return calculation. The targeted audit then examines 109 unique question-asset pairs selected from macroeconomic, geopolitical, regulatory, process, and non-standard contracts. Primary contracts are retained. Contracts that are circular, resolve from the mapped stock price, have unusable endpoints, or do not define an equity exposure are excluded. Process-only questions and contracts requiring an unavailable multi-contract probability transformation are marked secondary or quarantined rather than forced into H1.

Table 1 reconciles the data flow. The audit is targeted rather than universe-complete: 1,148 rows pass through deterministic rules without manual question-level review. This boundary is recorded because semantic validity, not only sample size, determines the credibility of the expectation estimate.

4 H1 Expectation Design

4.1 Estimand and execution

For each valid candidate, the primary net return is

$$r_i^{net} = \frac{P_{i,T_e-1} - P_{i,entry}}{P_{i,entry}} - \frac{C_i}{N_i},$$

Table 1: Candidate cleaning reconciliation.

Stage	Rows
Raw candidate rows	1,293
Exact duplicate excess removed	36
Rows after exact collapse	1,257
Audited unique question-asset pairs	109
Quarantined pending transformation	54
Secondary-only	11
Excluded or endpoint-invalid	10
Primary cleaned output	1,182

where P is the stored equity close, $N_i = \$10{,}000$ is fixed entry notional, and C_i contains round-trip commissions, regulatory fees, and 5 basis points of slippage per transaction leg. Whole shares are used. Every candidate is evaluated independently; portfolio capital, concurrency, Kelly sizing, and event-priority allocation are absent from H1.

The primary hypothesis is

$$H_1: \quad \mu_O = \mathbb{E}\left[r_i^{net} \mid i \in O_{clean}\right] > 0.$$

The expectation is reported at three levels. Candidate observations retain every valid question-asset row. Symbol-day collapse retains one observation for the same symbol and entry day. Economic-event aggregation first combines sibling contracts and repeated assets belonging to the same underlying event, then weights events equally. The event view prevents a dense question family from being mistaken for independent evidence.

4.2 Eligibility provenance

The eligibility parameters (strong and floor entry thresholds, confirmation days, probability-surge cap, and run-up cap) are not fixed a priori. They are the walk-forward policy schedule fitted by the portfolio optimizer of Section 6 (the all-treatment SPY arm). Each fold's policy is fitted only on candidates whose outcomes completed before that fold began, so every 2026 observation is evaluated under parameters that were point-in-time legal at its own signal date; however, the schedule is refit as the test period progresses, and observations before the first fold reuse the first fold's policy retrospectively. Two consequences are stated plainly. First, the full-sample H1 estimate is descriptive evidence about the opportunity set under the system's own selection rule, not an independent confirmatory hypothesis test. Second, the January–June 2026 subsample is the closest the data come to out-of-sample: its eligibility decisions use no future outcomes, but its policies were refit within the period and the period itself has been repeatedly inspected during research.

4.3 Inference and sensitivity analysis

An iid one-sided t-test is shown only as a reference. The primary uncertainty analysis resamples economic-event clusters. Additional cluster or block schemes use symbol, entry week, and entry month. Each bootstrap uses 20,000 replications with fixed seeds. Reported p-values are null-centered and one-sided ($p = (\#\{\text{null mean} \geq \text{observed}\} + 1)/(B + 1)$); intervals are ordinary uncentered percentile intervals of the resampled mean. We also report medians and positive-return frequencies because the mean can be positive in a right-skewed distribution even when close to half of observations win.

Sensitivity tests multiply modeled costs by up to three, shift entry to the next stored equity close, remove the largest positive observations, symmetrically trim both tails, and recompute estimates by calendar month and event family. A separate benchmark-relative analysis (Section 5.4) compares every trade against a matched benchmark position. These tests do not create new hypotheses; they measure how much of the central estimate is attributable to execution assumptions, shared calendar shocks, common market exposure, and the upper tail.

5 H1 Results

5.1 Central estimate

Of 1,182 cleaned candidates, 934 satisfy the probability entry rules. Eight hundred eighty-seven have a valid entry before the final eligible pre-event close; 47 enter too late or lack a usable price pair. The rejected remainder comprises 121 candidates below the entry thresholds, 108 exceeding the probability-surge cap, 12 without a clean signal side, and 7 exceeding the run-up cap. Table 2 reports the central results. Candidate observations have mean net return +1.208%, median +0.188%, and positive-return frequency 51.97% across 750 economic events, 387 symbols, 46 entry weeks, and 16 entry months. The event-cluster bootstrap interval is $[+0.263\%, +2.200\%]$ with one-sided $p = 0.0087$.

The result weakens as dependence becomes coarser. The candidate-symbol interval is $[-0.184\%, +2.776\%]$ with $p = 0.0652$, the entry-week block interval is $[-0.185\%, +2.834\%]$ with $p = 0.0678$, and the entry-month block interval is $[-0.630\%, +3.651\%]$ with $p = 0.1427$. After equal-weight event aggregation, the mean is +0.631% and the median is +0.013%; its month-block interval is $[-0.417\%, +2.407\%]$ with $p = 0.1935$. Figure 2 makes the dependence assumption visible rather than selecting one favorable standard error.

Restricting to the 528 candidates entered during January–June 2026 (point-in-time eligibility; Section 4.2) gives mean +2.038% and median +0.557%, with event-cluster $p = 0.0006$ but month-block $p = 0.1083$ on only six months.

5.2 Tail, cost, and timing sensitivity

The positive mean is upper-tail dependent. Removing the largest one percent of candidate returns lowers the candidate mean to +0.662%; removing the largest five percent makes it −0.260%. At the event level, removing the largest one percent reduces the mean from +0.631% to +0.029% ($p = 0.4523$), and removing the largest five percent produces −0.715%. Symmetric trimming is less destructive: a symmetric five-percent trim leaves candidate mean +0.584% and event mean +0.070%. This pattern is consistent with a broadly dispersed but strongly right-skewed payoff distribution.

At three times modeled transaction costs, the candidate mean remains +0.954% ($p = 0.0026$), although its median becomes slightly negative. Repricing entry at the next stored equity close yields candidate mean +1.229% and event mean +0.550%; the event one-sided p-value is 0.0550. Thus the headline is not eliminated by a one-close delay, but daily data cannot establish exact intraday tradability or after-hours sequencing.

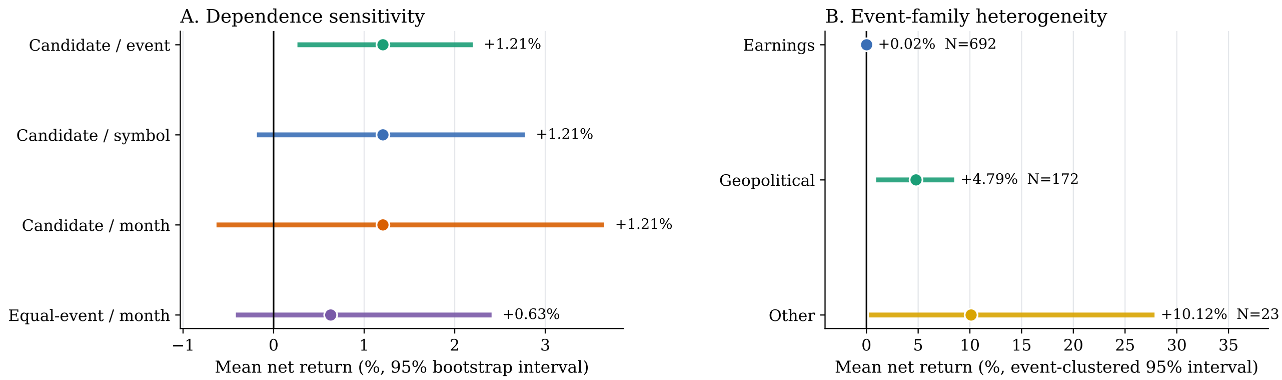

Figure 2: Cleaned H1 estimates. Panel A holds the candidate mean fixed while changing the resampling unit; the equal-weight event estimate is shown separately. Panel B is descriptive event-family heterogeneity, with candidate-weighted means and event-clustered intervals.

Table 2: Cleaned H1 expectation. All returns are net of modeled costs. Bootstrap p-values are null-centered and one-sided under the stated dependence unit; intervals are uncentered percentile intervals (20,000 replications).

Level and dependence unit	N	Mean	Median	Positive	95% bootstrap CI	p_{boot}
Candidate / economic event	887	+1.208%	+0.188%	51.97%	[+0.263, +2.200]%	0.0087
Candidate / symbol	887	+1.208%	+0.188%	51.97%	[−0.184, +2.776]%	0.0652
Candidate / entry month	887	+1.208%	+0.188%	51.97%	[−0.630, +3.651]%	0.1427
Symbol-day collapsed	786	+0.568%	−0.057%	49.62%	–	0.0611
Equal-weight event / entry month	750	+0.631%	+0.013%	50.27%	[−0.417, +2.407]%	0.1935

5.3 Polarity and event-family heterogeneity

A controlled ablation treats every YES probability as bullish while leaving the cleaned candidate file and H1 policy unchanged. That version produces 882 observations with mean +1.272% and median +0.229%, compared with 887 observations, mean +1.208%, and median +0.188% under resolved polarity. Therefore polarity does not improve aggregate return in this historical sample. Its contribution is semantic validity: it prevents logically inverted signals and permits contracts with favorable NO outcomes to be represented correctly. The 43 selected NO-bullish observations have mean −2.100% and median −2.864%, so that subgroup requires prospective review rather than a performance claim.

Event-family results are more informative for candidate choice. Earnings observations ($N = 692$) have mean +0.020% with event-clustered interval [−0.476%, +0.521%]. Geopolitical observations ($N = 172$, only 37 economic events) have mean +4.794%, median +2.443%, positive-return frequency 61.63%, and event-clustered interval [+0.936%, +8.485%]. The residual other group has mean +10.118% but only 23 observations and a very wide interval. These subgroup results were not separately preregistered. They motivate family-aware allocation, but do not establish that geopolitical classification is causal.

5.4 Benchmark-relative inference

The raw expectation could reflect common market exposure during a rising sample period. Each valid trade is therefore matched to an equal-notional benchmark trade over the identical entry and exit dates, with commissions, fees, and slippage modeled independently on both legs. Two controls are used: SPY for all 887 observations, and the candidate's actual sector ETF for the 680 observations with a real sector mapping (ETF-mapped geopolitical and macro candidates have no sector assignment and are excluded from the sector control rather than proxied by SPY).

Table 3: Benchmark-relative H1 inference. Each stock trade is matched to an equal-notional benchmark trade over the identical entry and exit dates, with costs modeled independently on both legs. No interval excludes zero.

Benchmark	Level / dependence	N	Mean	95% CI	p
SPY	Candidate / event cluster	887	+0.914%	[−0.116, +2.013]%	0.0508
SPY	Candidate / month block	887	+0.914%	[−0.899, +3.404]%	0.2036
SPY	Symbol-day / event cluster	786	+0.223%	[−0.453, +0.975]%	0.2641
SPY	Equal-event / event bootstrap	750	+0.430%	[−0.218, +1.174]%	0.1138
Sector ETF	Candidate / event cluster	680	+0.191%	[−0.393, +0.904]%	0.2644
Sector ETF	Candidate / month block	680	+0.191%	[−0.432, +1.459]%	0.3082
Sector ETF	Symbol-day / event cluster	680	+0.191%	[−0.385, +0.911]%	0.2662
Sector ETF	Equal-event / event bootstrap	678	+0.183%	[−0.398, +0.911]%	0.2727

Table 3 and Figure 3 report the result: the mean excess return is positive but no 95% interval excludes zero under any dependence scheme. Against SPY, the candidate mean excess is +0.914% with event-cluster $p = 0.0508$; symbol-day collapse reduces it to +0.223%, and the equal-weight event excess is +0.430% ($p = 0.1138$). Against sector ETFs the candidate excess is +0.191% ($p = 0.2644$). The

H1 evidence is therefore a positive raw conditional mean whose market-relative component is not statistically established in this sample; a substantial part of the raw headline coincides with market and sector movement over the same holding windows.

6 CEM Portfolio Implementation

6.1 Purpose and baseline policy

H1 evaluates every eligible observation at fixed notional. The portfolio experiment asks a different question: can one constrained policy select and size overlapping candidates while capital remains invested in either event positions or the benchmark? Starting capital is $100,000. Event positions use whole shares; idle cash above $100 is swept into fractional SPY or QQQ holdings. The simulator applies Interactive Brokers-style commissions, SEC fees on sales, and 5 basis points of slippage on each trade leg; per-trade profit is net of all four rotation legs (benchmark sell, asset buy, asset sell, benchmark re-buy).

CEM searches ten bounded policy parameters: ATR exit multiple $[1.5, 4.0]$, profit-lock activation $[2\%, 10\%]$, probability exit threshold $[0.45, 0.60]$, strong and floor entry thresholds, holding confirmation of one to five days, maximum probability surge $[0.20, 0.80]$, maximum pre-entry equity run-up $[2\%, 20\%]$, base position size $[6\%, 12\%]$, and maximum concurrency of 8–12 positions. The baseline fitness is

$$J(\pi) = \mathrm{Sharpe}_{daily}(\pi) - 0.30\,|\,\mathrm{MDD}_{\%}(\pi)|,$$

with the daily-equity Sharpe annualized by $\sqrt{252}$ and the maximum drawdown entering in percentage points. Each standard run uses six CEM iterations, population 20, elite fraction 25%, and a fixed seed. The robustness study varies both seed and CEM budget.

6.2 Treatments T1–T4

Each treatment changes one identifiable component of the implementation, following the portfolio branch in Figure 4. All treatment inputs are fixed constants or information available by the relevant decision time.

T1: realized-friction fitness. Let f_{fail} be the fraction of fitted trades whose gross PnL is below three times modeled transaction cost. T1 changes the CEM objective to $J(\pi) - 2f_{fail}$, preferring policies whose realized trades clear friction more consistently. It is not a live entry hurdle: the implementation has no point-in-time expected-return forecast, and T1 rejects no trade at execution time.

T2: label-complete walk-forward fitting. T2 constructs expanding three-month folds. A fold may fit only candidates with T_e strictly before the fold start; the following three-month candidate block is evaluated out of fold. At least two folds are required, with at least eight fit and eight evaluation candidates per fold. Every simulation is truncated at its fold horizon. T2 is therefore the main defense against unresolved-event labels and future price paths entering policy selection. Because the folds extend across the whole sample, T2 arms continue to refit inside the 2026 test period using only outcomes completed before each fold starts; they are online-refit schedules, not policies frozen at the end of 2025. Non-walk-forward arms (Baseline and T4 alone) fit once on the 423 candidates whose outcomes completed before 2026 and stay frozen.

T3: bounded half-Kelly sizing. After at least ten completed net trades, T3 estimates win probability and the mean win/loss ratio from the latest 30 completed trades. It applies one-half Kelly and clips the resulting allocation to 5–20%. Before ten completed trades, or when both wins and losses are not observed, it uses the CEM base size. Open positions and future outcomes never enter the sizing history.

T4: event-priority allocation. T4 changes the allocation of limited concurrent slots. Same-day candidates are ranked by event family (geopolitical, macro, earnings, other), then by favorable entry probability and pre-entry equity run-up clipped to $[-20\%, +20\%]$. Same-symbol same-day signals are collapsed into one position. If every slot is occupied by earnings positions, a geopolitical or macro candidate may replace the worst earnings position only when that position is below +3% net. T4 changes portfolio allocation, not the H1 observation set.

6.3 Chronological evaluation and optimizer robustness

The cleaned candidate file contains 427 training rows dated August 2024 through December 2025 and 755 rows dated January through June 2026; after the relevance filter, 1,148 candidates enter the experiment, of which 721 fall in the test window. Training eligibility depends on label completion: a candidate can enter a CEM fit only when its event endpoint is strictly earlier than the fit cutoff. Portfolio simulation in every fold is also truncated at that fold's decision horizon. No validation split is used anywhere in the pipeline: configuration families were declared in advance, and January–June 2026 is reported as one test period.

The first result matrix uses seed 42 and the standard 6×20 budget for five predeclared configurations: Baseline, T1+T2, T1+T2+T3, T4 alone, and all four treatments. Because a single optimizer draw can be lucky, a second experiment evaluates the flagship all-treatment arm across seeds 42–51 and budgets $(6, 20)$, $(10, 20)$, $(6, 30)$, and $(10, 30)$, and pairs each seed's all-treatment run against a Baseline run with the same seed at the standard budget. Distributions are summarized by median and interquartile range across seeds; no budget is selected or ranked by test performance.

7 CEM Results

7.1 Configuration matrix

Table 4 reports the seed-42, 6×20 matrix over the single January–June 2026 test period. All configurations have positive absolute return and positive excess return. SPY excess ranges from +16.98 to +22.35 percentage points; QQQ excess ranges from +4.98 to +9.77 points. T1+T2 has the strongest SPY result, while Baseline has the strongest QQQ result. The absence of a common winner is evidence against selecting one configuration from this matrix by its best terminal return. Return, Sharpe, and drawdown values were recomputed from the daily equity logs and match the reported results.

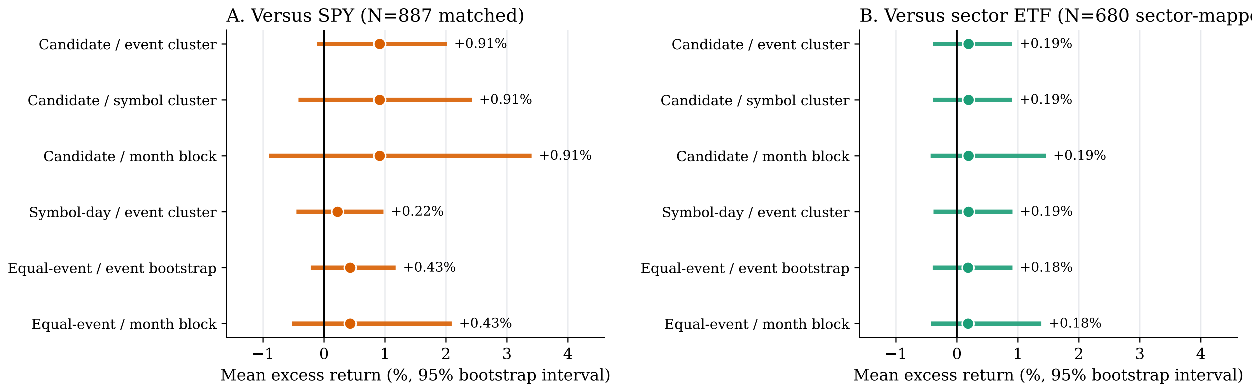

Figure 3: Benchmark-relative H1 inference: matched equal-notional benchmark trades over identical dates, after costs on both legs. Every 95% interval crosses zero.

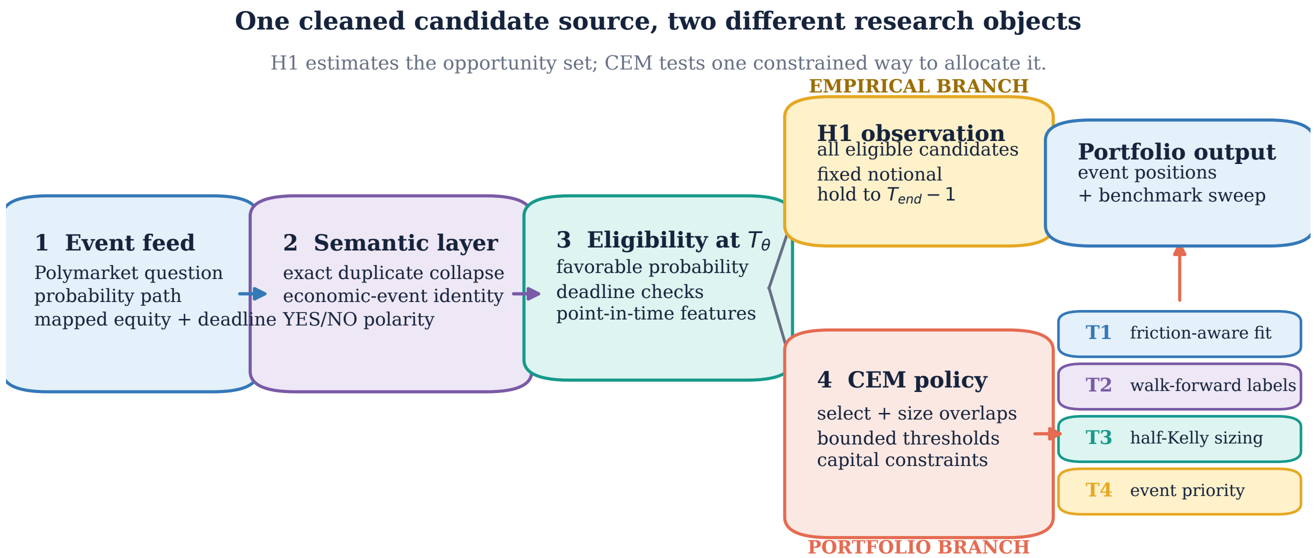

Figure 4: Research architecture. Cleaning and polarity create one eligible candidate source, after which the analysis deliberately separates two objects. H1 evaluates every eligible candidate at fixed notional through $T_{end} - 1$. The CEM branch selects and sizes overlapping candidates under capital constraints; T1–T4 modify fitness, chronology, sizing, and allocation, respectively. H1 output does not feed the portfolio results.

7.2 Effect of candidate cleaning

To isolate the cleaning change, the all-treatment arm is rerun on the original and audit-cleaned candidate universes using identical current code, seed 42, and 6×20 budget. On SPY, terminal return rises from 25.53% to 28.49%, a +2.96 point difference. On QQQ it rises from 22.29% to 24.61%, a +2.32 point difference. Five-day moving-block bootstrap tests of the cleaned-minus-original daily return give $p = 0.3371$ for SPY and $p = 0.3884$ for QQQ. Cleaning therefore improves these terminal point estimates, but the short matched path does not establish a systematic portfolio improvement.

7.3 Ten-seed and CEM-budget robustness

Figure 5 and Table 5 summarize 40 isolated flagship runs over the full test period. Median excess return is positive for every budget on both benchmarks: between +6.69 and +8.26 points on SPY (10 of 10 seeds positive in every budget) and between +3.89 and +6.44 points on QQQ (8–9 of 10 seeds positive). Budget medians differ by a few points with heavily overlapping interquartile ranges, so the evidence supports robustness of the candidate source to CEM random initialization rather than a preference for any optimizer budget. On the seed-42 flagship path, a 5-day moving-block bootstrap of daily excess return gives one-sided $p = 0.0317$ on SPY and $p = 0.3102$

Table 4: Audit-cleaned CEM test matrix over the single January–June 2026 test period (seed 42, six CEM iterations, population 20). Returns and excess are percentage points; benchmark buy-and-hold returned +8.52 (SPY) and +17.59 (QQQ) points.

	SPY benchmark					QQQ benchmark				
Configuration	Return	Excess	MaxDD	Sharpe	Trades	Return	Excess	MaxDD	Sharpe	Trades
Baseline	30.27	21.75	−6.67	3.30	225	27.36	9.77	−6.36	2.93	223
T1+T2	30.87	22.35	−9.52	3.41	191	24.63	7.03	−7.51	2.52	189
T1+T2+T3	25.50	16.98	−7.76	2.99	197	22.57	4.98	−11.43	2.10	181
T4 GeoPriority	27.27	18.75	−8.33	2.89	197	23.58	5.99	−6.74	2.54	211
T1+T2+T3+T4	28.49	19.97	−8.72	3.22	184	24.61	7.02	−10.38	2.20	185

on QQQ; given a single, short, and repeatedly inspected 111-day window, we treat this as descriptive rather than confirmatory.

7.4 Do the treatments beat the baseline?

Table 6 answers the comparison the matrix cannot: paired on the same seed at the standard 6×20 budget, the all-treatment configuration *underperforms* the plain CEM baseline. The median excess-return difference is −7.43 points on SPY (0 of 10 seeds favor the treatments) and −3.44 points on QQQ (3 of 10). The treatments trade less and pay lower transaction costs, but their median Sharpe is also lower. The honest conclusion is that the four treatments produce a profitable historical implementation without improving on the baseline policy search; their value is architectural (label-complete chronology, bounded sizing, explainable allocation) rather than performance-based in this sample.

8 Discussion

The primary empirical result is a positive observed conditional mean in the cleaned Polymarket-identified opportunity set. The candidate estimate is economically large relative to modeled cost and remains positive after event-cluster resampling. The same data do not establish a positive population expectation under every reasonable dependence model: symbol, week, and month clustering cross zero, the equal-event month-block interval crosses zero, and the event estimate is highly sensitive to removal of the upper tail. Most importantly, no benchmark-relative interval excludes zero, so the market-adjusted component of the headline is not statistically established. The appropriate conclusion is a positive historical conditional expectation with limited effective sample size and substantial concentration, not a universal return law and not demonstrated market-relative excess.

The strongest additional result is heterogeneity. Earnings questions contribute approximately zero mean return, while the geopolitical subset is positive in both mean and median and retains a positive event-clustered interval on only 37 economic events. This supports stricter candidate selection and motivated T4's family-aware allocation. It does not prove that geopolitical markets contain private information: a common public-news process, commodity exposure, recurring symbols, or favorable months can generate the same pattern, and the paired portfolio comparison shows the allocation rule did not convert the pattern into excess portfolio performance.

Polarity and cleaning serve different scientific roles. Polarity makes the signal economically interpretable; its value cannot be judged only by whether the historical mean rises. Cleaning defines the estimand by removing duplicates and invalid mappings. The matched CEM comparison is directionally favorable to cleaning but statistically inconclusive. Future candidate ingestion should therefore apply the rules prospectively: immutable candidate versions, exact duplicate collapse, explicit economic-event identifiers, polarity with reason codes, and quarantine of contracts requiring unavailable transformations.

For live use, the defensible role of Polymarket is a continuous confirmation variable. Probability level, direction, revisions, and deadline proximity can indicate when the crowd increasingly supports the economically favorable state. Those variables can inform candidate eligibility and exits without assuming minute-level trading or an information-diffusion lag. The current daily backtest cannot evaluate after-hours and weekend probability changes precisely, even though those periods may matter for large losses. A prospective hourly implementation should preserve the same semantic rules and evaluate only data observable at each decision time.

The principal limitations are the two-year venue history, one prediction-market source, a targeted rather than universe-complete semantic audit, daily historical equity decisions, eligibility parameters inherited from the portfolio optimizer rather than fixed a priori, and a short 2026 portfolio window. Repeated research iterations also mean that the 2026 period is not a pristine untouched holdout, and the walk-forward arms refit within it. The seed/budget protocol addresses optimizer-initialization luck, but only future data can provide a genuinely prospective test.

9 Conclusion

After exact duplicate collapse and targeted semantic cleaning, 887 valid Polymarket-linked candidate observations have mean net equity return +1.208% between signal eligibility and the final eligible pre-event close. Equal-weight economic events have mean +0.631%. Event-cluster inference supports a positive candidate mean, while symbol and calendar dependence and upper-tail removal materially weaken the result, and matched SPY and sector-ETF controls leave no interval that excludes zero. Geopolitical observations are the strongest descriptive family; earnings observations are centered near zero.

A fully specified CEM implementation demonstrates that the candidate source can be converted into profitable historical portfolios under transaction costs and capital constraints, and that this holds

Table 5: Flagship T1+T2+T3+T4 distribution across ten seeds over the full January–June 2026 test period. Entries are median [25th, 75th percentile] excess return in percentage points, the count of seeds with positive excess, and the median maximum drawdown. No budget is selected or ranked by these results.

	SPY			QQQ		
Budget	Excess median [IQR]	> 0	MaxDD	Excess median [IQR]	> 0	MaxDD
6×20	+8.26 [+4.31, +9.63]	10/10	−8.74	+3.89 [+2.21, +7.25]	8/10	−9.26
6×30	+6.69 [+6.15, +8.37]	10/10	−8.73	+5.65 [+3.11, +7.69]	9/10	−9.48
10×20	+7.76 [+7.35, +10.52]	10/10	−8.62	+4.65 [+1.33, +5.96]	8/10	−9.39
10×30	+6.97 [+5.77, +9.06]	10/10	−8.47	+6.44 [+4.15, +7.23]	9/10	−9.54

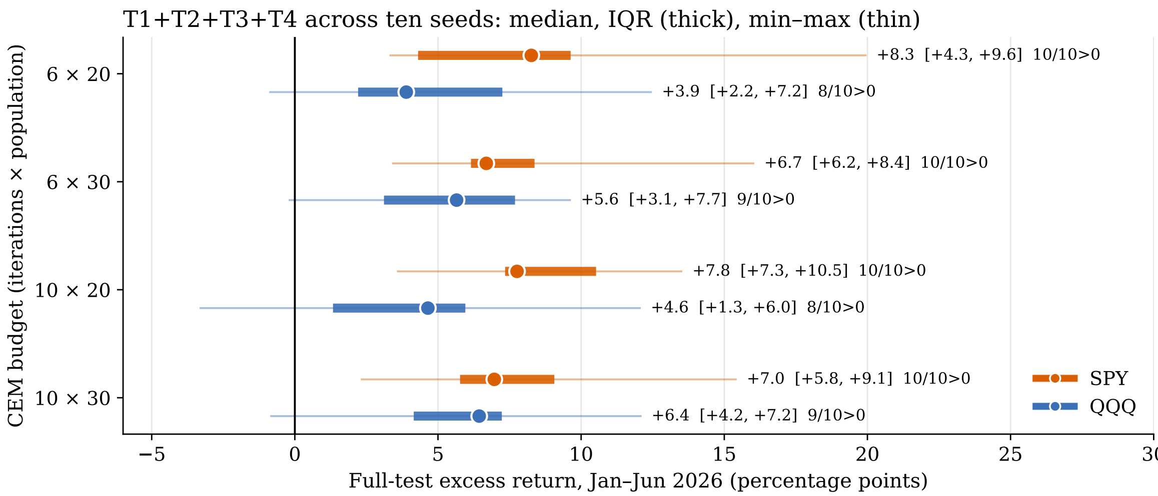

Figure 5: Flagship T1+T2+T3+T4 optimizer robustness across ten seeds and four CEM budgets over the full January–June 2026 test period. Medians are positive throughout; no budget is selected from these results.

Table 6: Paired comparison at the standard 6×20 budget: T1+T2+T3+T4 minus Baseline on the same seed and benchmark (ten seeds each). Entries are median [IQR] differences over the full test period; the last column is the share of seeds where the treated configuration has higher excess return.

	ΔExcess [IQR]	ΔSharpe	ΔMaxDD	ΔCost ($)	Wins
SPY	−7.43 [−12.39, −3.92]	−0.92	−0.16	−2,032	0%
QQQ	−3.44 [−7.87, +1.06]	−0.55	−1.00	−1,620	30%

across ten optimizer seeds and four search budgets on one contiguous January–June 2026 test period. It also shows, honestly, that the four studied treatments do not outperform the plain baseline in seed-paired comparisons, and that the short and repeatedly inspected test period is insufficient for a stable prospective-profitability claim.

The research contribution is therefore narrower and more defensible than information diffusion. Polymarket supplies continuous, timestamped crowd-belief variables that can identify event relevance, encode the favorable outcome through polarity, and confirm entry or exit conditions for selected equities. The positive conditional expectation shows that this information source is empirically useful in the observed sample. Establishing why it works, whether it beats simple market exposure, and whether it persists prospectively requires new events rather than additional optimization of the same history.

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